

Jiayu Kamessi Zhao

MIT Operations Research Center (ORC)
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APPOINTMENT Assistant Professor July 2026 (Expected)
Operations, Information, and Technology Area, Graduate School of Business
Stanford University, Palo Alto, CA

Postdoctoral Researcher July 2025 - May 2026
Supply Optimization Team
Waymo, Mountain View, CA

EDUCATION Massachusetts Institute of Technology (MIT), Cambridge, MA 2020 - 2025
Ph.D. in Operations Research
GPA: 5.0/5.0 Advisor: Daniel Freund

Columbia University, New York, NY 2016 - 2020
B.S. in Operations Research
Summa Cum Laude

RESEARCH INTERESTS My research studies how two-sided service platforms, via algorithm and market designs, incentivize agents' flexibility to enhance operational efficiency. Through the lens of flexibility designs, I aim to (i) manage incentives across both supply and demand sides of platforms to improve matching efficiency, and (ii) align agents' incentives to adopt new technologies, such as deploying autonomous vehicles on mobility platforms. In tackling these challenges, I apply tools from game theory and stochastic decision-making to transportation, e-commerce, and other platform applications.

PUBLICATIONS AND PREPRINTS *"The Value of a Little Flexibility in Stable Matching,"* with Daniel Freund and Sébastien Martin. Available at SSRN 6208698, 2026.

"Two-sided Flexibility in Platforms," with Daniel Freund and Sébastien Martin.
Major Revision at **Operations Research**, 2026. Available at Arxiv 2404.04709.

- **MIT ORC Best Student Paper Award, 2024**
- Accepted for presentation at 2024 MSOM Conference
- Accepted for presentation at 2024 Revenue Management & Pricing Conference
- Accepted for presentation at 2024 Marketplace Innovation Workshop

"On the Power of Delayed Flexibility: Balls, Bins, and a Few Opaque Promotions,"
with Daniel Freund and Chamsi Hssaine.

Major Revision at **Operations Research**, 2026. Available at Arxiv 2306.01968.

- Accepted for presentation at 2025 Marketplace Innovation Workshop

"On the Supply of Autonomous Vehicles in Platforms," with Daniel Freund and Ilan Lobel. Accepted by **Manufacturing & Service Operations Management**.

- **Twenty-Fifth ACM Conference on Economics and Computation (EC'24)**
- Accepted for presentation at **2024 MSOM Supply Chain Management SIG**
- Accepted for presentation at 2023 Revenue Management & Pricing Conference
- Accepted for presentation at 2023 Marketplace Innovation Workshop

"Overbooking with Bounded Loss," with Daniel Freund. 2022.
Mathematics of Operations Research 48(3): 1344-1363.

- **Twenty-Second ACM Conference on Economics and Computation (EC'21)**
- Accepted for presentation at 2021 Marketplace Innovation Workshop

WORK IN PROGRESS

“*Stay, Reposition, or Recharge: Battery-Aware Operation of Robotaxi Fleets*,” with Saif Benjaafar and Philipp Afèche.
 “*Robotaxis as Capacity Expansion: Certification-Based Pricing and Matching*,” with Huiwen Jia, Xuchen Liu and Siqian Shen.

EXPERIENCES

Uber Technologies, Inc., San Francisco, CA Summer 2023 & 2024
Applied Scientist Intern Supervisors: Ali Sadighian, Shaiza Qayyum, AJ Friend

- Uber Eats Eater Realtime Pricing Team (ERTP)
- Improved the convex optimization program for Uber Eats’ real-time pricing and incentive design; enhanced the real-time acceptance rate model and the chained supply model for mitigating surge demand

MIT ORC, Cambridge, MA 2020 - 2025
Research Assistant Supervisor: Daniel Freund

Simons Institute for the Theory of Computing, UC Berkeley, CA Fall 2022
 Participant of the Data-Driven Decision Processes Program

Columbia IEOR Department, New York, NY 2018 - 2020
Research Assistant Supervisors: Jay Sethuraman, Soulaymane Kachani.

TEACHING

UC Berkeley, Berkeley, CA
 Guest Lecture for *INDENG 150: Production Systems* (Undergrad) Fall 2025
 Guest Lecture for *INDENG 250: Production and Logistics Models* (PhD) Fall 2025

MIT, Cambridge, MA
 TA for *15.761 Operations Management* (MBA) Spring 2022, 2023, 2024
 TA for *15.S25 Experience in Operations Research* (PhD) Summer 2022

Columbia University, New York, NY
 TA for *Math 2030 Ordinary Differential Equations* (Undergrad) Fall 2018
 TA for *IEOR E3658 Probability for Engineers* (Undergrad) Fall 2018

SELECTED TALKS

“*The Value of a Little Flexibility in Stable Matching*”
 - INFORMS APS Conference, Jul 2025, Atlanta GA

“*On the Power of Delayed Flexibility: Balls, Bins, and a Few Opaque Promotions*”
 - Marketplace Innovation Workshop, May 2025, Virtual

“*Two-sided Flexibility in Platforms*”
 - INFORMS APS Conference, Jul 2025, Atlanta GA
 - University of British Columbia, Imperial College, London Business School, University College London, National University of Singapore, New York University, University of Toronto, Duke University, Stanford University, Columbia University
 - INFORMS Annual Meeting, Oct 2024, Seattle WA
 - Inaugural OM Rookiepalooza at Kellogg, Oct 2024, Evanston IL

- MIT ORC Fall Seminar Series, Nov 2024, Cambridge MA
- RMP Conference, Jul 2024, Los Angeles CA
- MSOM Conference, Jul 2024, Minneapolis MN
- Marketplace Innovation Workshop, May 2024, Virtual
- MIT LIDS Student Conference, Feb 2024, Cambridge MA
- Xi'an Jiaotong University, Jan 2024, Xi'an, China
- Shanghai University of Finance and Economics, Dec 2023, Shanghai, China

“On the Supply of Autonomous Vehicles in Platforms”

- MIT Operations Management Seminar, Oct 2024, Cambridge MA
- MSOM Conference Supply Chain Management SIG, Jun 2024, Minneapolis MN
- INFORMS Annual Meeting, Oct 2023, Phoenix AZ
- Cornell Young Researchers Workshop, Oct 2023, Ithaca NY
- MSOM Conference, Jun 2023, Montreal
- INFORMS Annual Meeting, Oct 2022, Indianapolis IN
- Marketplace Innovation Workshop, May 2023, Virtual

“Overbooking with Bounded Loss”

- EC'21, Jul 2021, Virtual
- INFORMS Annual Meeting, Oct 2021, Virtual
- ORC Student Seminar, Apr 2022, Cambridge MA

PROFESSIONAL SERVICES	Reviewer for <i>Management Science, Operations Research</i>	2025 - Present
	Reviewer for the ACM STOC Conference	2026
	Reviewer for the ACM EC Conference	2025
	Reviewer for the MSOM Conference	2025
	MIT ORC Seminar Series student coordinator	Fall 2023
	MIT Operations Management Seminar student co-organizer	2021 - 2023
HONORS AND AWARDS	Dan & Eva Roos Thesis Award	2025
	MIT ORC Best Student Paper Award	2024
	The Sebastian B. Littauer Award, Columbia University	2020
	Tau Beta Pi Honor Society, Columbia University (NY Chapter)	2018 - 2020
	The Dean's List, Columbia University,	2016 - 2020
	C.P. Davis Scholar, Columbia University	2016